SAHIL BASU

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Experience

Sugarlabs/musicblocks – Open Source Contributor Aug 2026 – Present
- Contributed 8+ merged pull requests to Sugar Labs' Music Blocks using JavaScript and Jest.
- Added tests and fixed bugs in contributor workflows.
- Worked with maintainers on code reviews, automated tests, and GitHub CI/CD pipelines.

TechGlaz Labs Pvt. Ltd. – Full Stack Developer Intern – Bhagalpur, India Jun 2026 – Jul 2026
Certificate
- Added 30+ production features and fixed 100+ bugs across application modules.
- Implemented role-based access control to prevent unauthorized access to application modules.
- Identified and resolved API performance bottlenecks.
- Refined MongoDB schemas and request validation to fix data inconsistencies.

Projects

MINEGOV AI — National Mining Governance & Telemetry Platform
Tech: React 19, Node.js, Express, PostgreSQL, WebSockets, Leaflet, Chart.js, Claudinary | GitHub
- Built a mining surveillance platform in React 19, Node.js, and PostgreSQL to manage operations across 8 coal subsidiaries.
- Built WebSocket ingestion for SCADA gas sensors (CH4/CO), JWT-based role access, and SHA-256 audit logs for statutory enforcement orders.
- Added Leaflet satellite maps for open-cast pit benchmarks and Chart.js dashboards with synchronized dark and light modes.

AI Career Navigator & Smart Student Hub
Tech: React.js, Firebase, Gemini API, REST APIs, WebRTC | GitHub
- Built an AI career guidance platform with Gemini LLM for mentoring, learning roadmaps, and career advice.
- Built an ATS resume analyzer using NLP to flag skill gaps and provide optimization metrics.
- Built a recruiter portal and dashboard with Firebase Auth and Firestore for profile management and PDF portfolio exports.

WellSenseAI
Tech: Python, FastAPI, LangChain, ChromaDB, Sentence Transformers, PyMuPDF, Pydantic, REST APIs | GitHub
- Built an asynchronous FastAPI RAG service to analyze drilling incidents, well-control events, telemetry data, and operational hazards.
- Indexed PDFs with PyMuPDF and LangChain chunking, then used 384-dimensional Sentence Transformers embeddings and ChromaDB for semantic search.
- Built REST APIs for indexing, retrieval, risk prediction, health monitoring, vector database statistics, and collection reset.
- Added CORS support for a React frontend, Swagger/ReDoc documentation, and automated end-to-end API tests.

Education

Guru Gobind Singh Indraprastha University (GGSIPU) – New Delhi, India 2028
B.Tech – Information Technology (3rd year) CGPA: 8.25

Technical Skills

Programming Languages	C, C++, JavaScript, Python
Web Technologies	HTML5, CSS3, React.js, Node.js, Express.js, RESTful APIs, Tailwind CSS
Databases	MongoDB, Mongoose, MySQL
Tools & Platforms	Git, GitHub, Postman, Firebase, Vercel, Netlify, VS Code, Render, Railway
Problem Solving	500+ problems on LeetCode, GFG and Codeforces, max rating: 1470

Achievements

- **1st Place Winner** TechSprint Hackathon — **Certificate**
- 3× Hackathon Finalist across national-level competitions.
- **Smart India Hackathon (SIH) 2026**: Selected at college level as a team member. **Certificate**